HOWARD (ZIYU) HAN

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA, United States 2024 – Present
Ph.D. in Human-Computer Interaction
Advisor: Nik Martelaro

Carnegie Mellon University, Pittsburgh, PA, United States 2021 – 2023
Master of Science in Computational Design
Thesis Committee: Daragh Byrne, Sarah Fox, Nikolas Martelaro

Tongji University, Shanghai, China 2016 – 2021
Bachelor of Engineering in Urban Planning

ONGOING PROJECTS

Community-Driven Public Space Robot Social Navigation and Communication Sep 2024 – Present
Lead PhD — Leading this NSF-funded project ($1.25M) with CMU Metro21 and the Urban Robot Foundation. Developing socially compliant behaviors for robot navigation and humanoid locomotion.
– Papers: CHI '24 [C5], ASSETS '23 [S3], HRI '26 LBR [S4]; under review at HRI '27 [C7] and IJRR [C4]
– Featured in CMU News, AAAS EurekAlert!, Tech Xplore, CMU SCS News, and TribLive

Human-Agent Interaction for More Adaptive and Contested Design Processes Jan 2025 – Present
Lead PhD — Developing modeling techniques and technical systems that help designers coordinate multiple LLM agents representing diverse modes of thinking to enhance adaptive design processes.
– Papers: HAI '26 [C6]; under review at CHI '27 [C8]

PUBLICATIONS

Conference & Journal Papers

[C8] *Under Review* **CHI '27**
Investigating How Designers Perceive the Timing and Content of Proactive AI-generated Feedback
Howard Han, Sirui Tao, Jingwen Pan, Seung Heon Song, Steven P. Dow, Nikolas Martelaro
Under review at ACM CHI Conference on Human Factors in Computing Systems, 2027

[C7] *Under Review* **HRI '27**
Learning Social Navigation Strategies and Challenging Scenarios through Public-engaged Simulation
Howard Han, Ying Zhang, Grace Yang, Yulin Chen, Aaron Steinfeld, Allan Wang, Sarah E. Fox, Nikolas Martelaro
Under review at ACM/IEEE International Conference on Human-Robot Interaction, 2027

[C6] **HAI '26**
Enacting Constructive Conflicts with AI Agents to Enhance Reconsideration among Novice Interaction Designers
Howard Han, Nikolas Martelaro
ACM International Conference on Human-Agent Interaction, 2026

[C5] **CHI '24**
Co-design Accessible Public Robots: Insights from People with Mobility Disability, Robotic Practitioners and Their Collaborations
Howard Han, Franklin Mingzhe Li, Alesandra Baca Vazquez, Daragh Byrne, Nikolas Martelaro, Sarah E. Fox
ACM CHI Conference on Human Factors in Computing Systems, 2024

[C4] *Under Review* **IJRR**
ACME: A Multi-Cultural, Multi-Embodiment Social-Navigation Dataset
Shashank Rao Marpally, Allan Wang, Atharva Ghotavadekar, . . . , **Howard Han**, . . . , Christoforos Mavrogiannis, Harold Soh (39 authors)
Under review at The International Journal of Robotics Research; arXiv:2607.21964, 2026

[C3] **CSCW '25**
WeAudit: Scaffolding User Auditors and AI Practitioners in Auditing Generative AI
Wesley Hanwen Deng, Claire Wang, **Howard Han**, Jason I. Hong, Kenneth Holstein, Motahhare Eslami
Proceedings of the ACM on Human-Computer Interaction (CSCW), 9(7), 2025
Best Paper Award

[C2] **CHI '25**
Raise Your Eyebrows Higher: Facilitating Emotional Communication in Social Virtual Reality Through Region-Specific Facial Expression Exaggeration
Xueyang Wang, Sheng Zhao, Yihe Wang, **Howard Han**, Xinge Liu, Xin Yi, Xin Tong, Hewu Li
ACM CHI Conference on Human Factors in Computing Systems, 2025

[C1] **JoDLA '21**
Evolution of Historical Urban Landscape with Computer Vision and Machine Learning: A Case Study of Berlin
Hui Tian, **Howard Han**, Weishun Xu, Xun Liu, Waishan Qiu, Wenjing Li
Journal of Digital Landscape Architecture, 6, 2021

Short Papers, Posters & Workshop Papers

[S4] *Late-Breaking Report* **HRI '26**
Enhanced Social Navigation Simulation for Community-Driven Participatory Evaluation of Sidewalk Robots
Howard Han, Ying Zhang, Allan Wang, Nikolas Martelaro
Companion Proceedings of the ACM/IEEE International Conference on Human-Robot Interaction, 2026

[S3] *Poster* **ASSETS '23**
The Robot in Our Path: Investigating the Perceptions of People with Motor Disabilities on Navigating Public Space Alongside Sidewalk Robots
Howard Han, Franklin Mingzhe Li, Nikolas Martelaro, Daragh Byrne, Sarah E. Fox
ACM SIGACCESS Conference on Computers and Accessibility, 2023

[S2] *Workshop Paper* **ASSETS '22**
Integrating Urban Accessibility into Human-Robot Interaction Evaluation
Howard Han, Sarah E. Fox, Nikolas Martelaro
The 2nd Annual Workshop on the Future of Urban Accessibility at ASSETS, 2022

[S1] *Poster* **IEEE VR '23**
Catch My Eyebrow, Catch My Mind: Examining the Effect of Upper Facial Expressions on Emotional Recognition for VR Avatars
Xin Yi, **Howard Han**, Xinge Liu, Yutong Ren, Xin Tong, Yan Kong, Hewu Li
IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), 2023

Thesis

[T1] **CMU '23**
Designing for Accessible Sidewalk Robots: Insights from People with Motor Disabilities
Howard Han
Master's Thesis, Carnegie Mellon University, 2023

RESEARCH EXPERIENCE

Research Assistant at CMU Human-Computer Interaction Institute — Sep 2023 – Present
Pittsburgh, PA

- Initiated and led a Human-Robot Interaction (HRI) project to evaluate how people with disabilities perceive the designs of sidewalk robots using simulation videos and co-design accessible public service robot technology, resulting in a CHI 2024 full paper [C5] and an ASSETS 2023 poster [S3].

Research Associate at CoALA Lab at CMU — Feb 2023 – Sep 2023
Advised by Prof. Ken Holstein and Prof. Motahhare Eslami — *Pittsburgh, PA*

- Co-led a Responsible AI project to understand how everyday users search for socially harmful behaviors in text-to-image systems and develop a pipeline to scaffold bias searching, resulting in a CSCW 2023 workshop paper and a CSCW 2025 full paper [C3] (Best Paper Award).

Research Assistant at Tsinghua Pervasive HCI Group — May 2022 – Feb 2023
Advised by Prof. Xin Yi and Prof. Xin Tong (Duke Kunshan University) — *Beijing, China*

- Led a social VR project to evaluate virtual avatar facial expressions and trained an ML model to augment VR communications by generating masked upper facial expressions, resulting in an IEEE VR 2023 poster [S1].

SKILLS

Technical Skills	Python, C, C#, JavaScript, Unity, PyTorch, HTML/CSS, R, React, Arduino, GIS, ROS
Design Skills	3D modeling, UI/UX, Photoshop, Video editing, Animations, Figma, Sketch
Research Skills	Qualitative coding, User studies, Interviews, Quantitative data analysis, Prototyping

AWARDS

- CMU Graduate Small Project Help (GuSH) Research Funding $\times$2 ($7,502)
- CMU Merit Scholarship ($20,000)
- CMU Computational Design Microgrant ($1,000)
- Challenge Cup National Third Prize (China's biggest student innovation award)

ACADEMIC SERVICE

DIS Student Volunteer	2023
DIS Reviewer	2023
IEEE VR Reviewer	2024